

Coulomb-metric point selection in interpolative separable density fitting

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The interpolative separable density fitting (ISDF) representation of the two-electron Coulomb integrals underpins low-scaling implementations of hybrid density-functional theory, many-body perturbation theory, and quantum-simulation resource estimation. The efficiency of the representation is governed by the number of interpolation points per orbital, $c_\mu = N_\mu/N_{\text{orb}}$, required to reach a target accuracy: auxiliary-space storage scales as $O(N_\mu^2)$ and dense auxiliary-space linear algebra as $O(N_\mu^3)$. Interpolation points are conventionally selected by a pivoted Cholesky factorization of the orbital-pair overlap matrix, an objective that weights all Fourier components of the pair densities equally and consequently over-resolves short-wavelength structure that the Coulomb interaction suppresses. Starting from the observation that the grid-space metric cancels exactly from the ISDF least-squares problem, so that any metric can influence the factorization only through the choice of interpolation points and through pair-space weights, we introduce three composable modifications of the point-selection procedure: (i) a separable weighting of the selection Gram matrix in the space of orbital pairs, (ii) point selection on Gaussian-filtered surrogate orbitals, and (iii) an exact Coulomb-metric re-ranking of an enlarged candidate pool generated by the overlap-metric factorization. None of these modifications alters the ISDF ansatz, the interpolation-vector equations, or any downstream consumer of the factorization, and each reduces to the conventional algorithm in a well-defined limit. For bulk silicon described by norm-conserving, ultrasoft, and projector augmented-wave pseudopotentials, we quantify the error in the Hartree, exact-exchange, and random-phase-approximation correlation energies as a function of c_μ [and find: summary of c_μ^* reductions, per scheme and composed]. The re-ranking step increases the cost of point selection by a factor $s \approx 2$ without changing its asymptotic scaling, and point selection remains a negligible fraction of the total cost of the calculation.

I. INTRODUCTION

Low-scaling formulations of electronic-structure methods beyond density-functional theory rely on compressed representations of the two-electron repulsion integrals. Among these, the interpolative separable density fitting (ISDF), or tensor hypercontraction (THC), representation factorizes the orbital-pair densities through a set of real-space interpolation points,

$$\rho_p(\mathbf{r}) \equiv u_i^*(\mathbf{r}) u_a(\mathbf{r}) \approx \sum_{\mu=1}^{N_\mu} \zeta_\mu(\mathbf{r}) \rho_p(\mathbf{r}_\mu), \quad (1)$$

and thereby reduces the four-index Coulomb tensor to products of two-index quantities [cite: Lu–Ying 2015; Hu–Lin–Yang 2017; Parrish–Hohenstein–Martínez THC; CoQui]. The representation has been employed in Hartree–Fock and hybrid-functional exchange, Møller–Plesset and random-phase-approximation (RPA) correlation energies, coupled-cluster theory, GW and related many-body perturbation frameworks, and in resource estimates for fault-tolerant quantum simulation [cites]. In all of these applications the auxiliary (interpolation-point) index carries the factorization: every object stored on that index requires $O(N_\mu^2)$ memory, and every factorization or inversion in the auxiliary space requires $O(N_\mu^3)$ operations. The proportionality constant $c_\mu = N_\mu/N_{\text{orb}}$ at fixed target accuracy is therefore the central figure of merit of the method; a reduction of c_μ from 10 to 6 corresponds to a factor of 2.8 in auxiliary-space storage

and 4.6 in dense auxiliary-space linear algebra, uniformly across applications.

In the standard algorithm the interpolation points are selected by a pivoted Cholesky factorization of the Gram matrix of orbital-pair densities, which greedily minimizes the L^2 error of the fitted pair densities on the real-space grid [cite]. This objective is optimal for the pair densities themselves but not for the quantities the factorization is used to compute: every consumer of the representation contracts it against the Coulomb kernel $v(\mathbf{G}) = 4\pi/|\mathbf{G}|^2$, or a screened variant thereof, which suppresses short-wavelength errors by two powers of $|\mathbf{G}|$. Points selected in the overlap metric therefore allocate resolution to features of the pair densities that contribute negligibly to Coulomb matrix elements. [One or two sentences positioning against prior work on alternative selection schemes (centroidal Voronoi tessellation, machine-learned selection), with citations.]

In this work we adapt the point selection—and only the point selection—to the Coulomb metric. The analysis begins from an exact structural property of the ISDF least-squares problem, established in Sec. II A: the interpolation vectors are independent of any grid-space metric, which cancels identically from the normal equations. All freedom therefore resides in the choice of interpolation points and in pair-space weights. We exploit this freedom through three independent and mutually compatible modifications of the selection procedure (Sec. II), all of which reuse the existing pivoted-Cholesky machinery, are controlled from the program input, and reduce to the standard algorithm in well-defined

limits. Section III quantifies the resulting errors in the Hartree, exact-exchange, and RPA correlation energies of bulk silicon for norm-conserving, ultrasoft, and projector augmented-wave (PAW) pseudopotentials.

II. THEORY

A. Exact cancellation of the grid metric

Let the pair densities be collected on a real-space grid in the matrix $Z \in \mathbb{C}^{N_g \times N_{\text{pair}}}$, $Z_{gp} = \rho_p(\mathbf{r}_g)$, and let $\Theta = Z[I, :]$ denote its restriction to a set of interpolation points $I = \{\mathbf{r}_\mu\}$. The ISDF ansatz (1) reads $Z \approx \zeta \Theta$ with $\zeta \in \mathbb{C}^{N_g \times N_\mu}$. Consider the weighted least-squares functional

$$E = \text{Tr}[(Z - \zeta \Theta) W (Z - \zeta \Theta)^\dagger v], \quad (2)$$

with an arbitrary Hermitian positive-definite grid metric v and a diagonal pair-space weight $W = \text{diag}(w_p)$, $w_p \geq 0$. The stationarity condition $v(Z - \zeta \Theta) W \Theta^\dagger = 0$, together with the invertibility of v , yields

$$\zeta = Z W \Theta^\dagger (\Theta W \Theta^\dagger)^{-1}. \quad (3)$$

The grid metric cancels identically, whereas the pair weight does not. The cancellation has a simple structural origin: ζ carries an unconstrained grid index, so the residual is driven orthogonal to the row space of Θ at each grid point independently, and no reweighting across grid points can improve upon this. Three consequences inform all subsequent developments. First, the interpolation vectors produced by the standard algorithm are already optimal in *any* grid metric for the given point set; there is nothing to correct in the solve. Second, no alternating-optimization or higher-scaling refinement of ζ at fixed I can lower the error in any grid metric. Third, a metric can enter the factorization only through the selection of I and through the pair weights W ; both mechanisms are inexpensive, and they define the three schemes introduced below.

B. The selection functional

With ζ eliminated through Eq. (3), the fit error in the metric v becomes a function of the point set alone,

$$E_v(I) = \text{Tr}[Q_\perp M], \quad M = Z^\dagger v Z, \quad (4)$$

where $Q_\perp$ projects onto the orthogonal complement, in pair space, of the rows of Θ . Greedy minimization of Eq. (4) is equivalent to pivoted Cholesky factorization of

$$K = Z M Z^\dagger = C v C, \quad C = Z Z^\dagger, \quad (5)$$

in which $C_{gg'} = A_{gg'} B_{gg'}^*$ inherits the Hadamard structure of a product of orbital Gram matrices, $A_{gg'} =$

$\sum_i u_i(\mathbf{r}_g) u_i^*(\mathbf{r}_{g'})$; this structure renders the diagonal and individual columns of C computable in $O(N_g N_{\text{orb}})$ operations. The conventional selection corresponds to the replacement $M \rightarrow \mathbb{K}$, i.e., to pivoting on C itself. Direct pivoting on K is not affordable: the Coulomb kernel is a convolution and does not commute with the Hadamard structure, so that a single column of K costs $O(N_g N_i N_a)$ and a complete selection $O(N_g N_{\text{orb}}^3)$. The three schemes below approximate or realize the v -weighted objective at essentially the cost of the standard algorithm.

C. Pair-space weighting

The pair weight W of Eq. (2) survives the elimination of ζ and modifies the selection functional directly. A general weight w_{ia} would destroy the Hadamard factorization that makes the selection affordable, but any weight expressible as a short sum of separable nonnegative terms,

$$w_{ia} = \sum_{p=1}^{N_L} c_p f_p(\varepsilon_i) g_p(\varepsilon_a), \quad c_p, f_p, g_p \geq 0, \quad (6)$$

preserves it at N_L times the cost: each term contributes a Gram matrix of scaled orbitals, and the weighted selection matrix $C^w = \sum_p c_p A^{(p)} \odot B^{(p)*}$ remains positive semidefinite by the Schur product theorem, so the pivoted factorization retains its numerical robustness. In practice the weights are functions of the single-particle energies. Energy-denominator weights, $w_{ia} = (|\varepsilon_i - \mu| + |\varepsilon_a - \mu| + \eta)^{-1}$, which reduce to the familiar $1/(\varepsilon_a - \varepsilon_i + \eta)$ for occupied-virtual pairs, admit a Laplace expansion into the form of Eq. (6) with all coefficients nonnegative and no numerical overflow; quadrature orders $N_L = 4$ –6 suffice, since the pivot ordering is a discrete choice far more forgiving than a continuous fit. Beyond fixed functional forms, Eq. (6) defines a design space: a rank-1 weight ($N_L = 1$) costs no more than the unweighted selection, and the parameters of a simple family $f(\varepsilon)$ can be optimized directly against the error in the target observables. The weight is applied to the point selection only; the interpolation vectors are always obtained from Eq. (3) with $W = \mathbb{K}$, so a single point set can serve several consumers. Unweighted selection depends only on the span of the orbital set and is therefore invariant under unitary rotations within it; energy-dependent weights forfeit this invariance, and the point set should then be selected once and reused.

D. Filtered-orbital surrogate

The second scheme performs the point selection on attenuated surrogate orbitals whose plane-wave coefficients are damped by a Gaussian,

$$\tilde{c}_{n\mathbf{k}}(\mathbf{G}) = c_{n\mathbf{k}}(\mathbf{G}) e^{-\alpha|\mathbf{k}+\mathbf{G}|^2/G_c^2}, \quad (7)$$

where G_c denotes the wave-function cutoff. The selection Gram matrix is assembled from the $\tilde{u}$, the pivoted factorization selects the points, and Eq. (3) is subsequently solved with the *unfiltered* orbitals; only the point set is retained from the surrogate. Because the Fourier transform of a pair density is a convolution of the transforms of its factors, filtering both factors attenuates the product at an effective width reduced by $\sqrt{2}$ and with a Gaussian rather than $1/G^2$ profile; the scheme is therefore a surrogate for, rather than an implementation of, the Coulomb metric. Its appeal is practical: the cost is a single additional Fourier-transform sweep during selection, and the limit $\alpha = 0$ recovers the standard algorithm exactly.

E. Coulomb-metric re-ranking of a candidate pool

The third scheme realizes the Coulomb-metric objective exactly on a restricted search space. The standard (or weighted, or filtered) pivoted factorization is first carried to an enlarged rank $N_1 = \lceil sN_\mu \rceil$ with $s \in [1.5, 3]$, producing a candidate pool P and the Cholesky factor $L \in \mathbb{C}^{N_g \times N_1}$. Because the Schur complement vanishes identically on eliminated pivots, the restriction of the Gram matrix to the pool is reproduced exactly by the factor, $C[P, :] = \ell L^\dagger$ with $\ell = L[P, :]$ lower triangular. Consequently,

$$K_{PP} = \ell (L^\dagger v L) \ell^\dagger = Y^\dagger Y, \quad Y = R \ell^\dagger, \quad (8)$$

where $R(\mathbf{G}, j) = \sqrt{\bar{v}(\mathbf{G})} \hat{L}(\mathbf{G}, j)$ is assembled from the Fourier transforms of the columns of L and a metric multiplier $\bar{v}(\mathbf{G})$, for which we consider the bare kernel, a Gaussian-attenuated kernel, and Brillouin-zone averages thereof. A dense pivoted Cholesky factorization of the $N_1 \times N_1$ matrix K_{PP} then re-ranks the pool in the Coulomb metric, and the leading N_μ points are retained. The Coulomb-optimal subset is sought only within the overlap-selected pool; this restriction is what renders the procedure affordable, at a cost of approximately a factor s on the selection step with no change in asymptotic scaling. The limits $s = 1$ and $\bar{v} = 1$ reproduce the pool and its ordering exactly.

F. Implementation

The three schemes are implemented in the CoQui code [cite] within the existing pivoted-Cholesky selection kernels and are controlled entirely from the program input (Table I); all defaults reproduce the conventional algorithm to machine precision. The filter of Eq. (7) is applied to the plane-wave coefficients from which the selection Gram matrix is assembled, while a second, unfiltered set of orbital values supplies the collocation matrices $\varphi_n(\mathbf{r}_\mu)$ returned to the solver; the surrogate therefore cannot contaminate the fitted representation. The pair weights enter the Gram diagonal and the gathered pivot

TABLE I. Input parameters controlling the modified point selection (THC/ISDF input block of CoQui). Default values reproduce the conventional overlap-metric algorithm.

parameter	default	role
<code>isdf_filter_alpha</code>	0	filter exponent α , Eq. (7)
<code>isdf_pair_weight</code>	none	pair weight; $\text{gap} = \text{selects } w_{ia} = (\varepsilon_i - \mu + \varepsilon_a - \mu + \eta)^{-1}$
<code>isdf_laplace_terms</code>	4	Laplace expansion order N_L
<code>isdf_eta</code>	0.01	regularization η (Ha)
<code>isdf_metric</code>	l2	re-ranking metric (bare, attenuated)
<code>isdf_pool_factor</code>	2	pool inflation factor s
<code>isdf_metric_qavg</code>	true	Brillouin-zone average of $\bar{v}$
<code>isdf_metric_gcut</code>	0	metric reciprocal-space truncation (Ha)
<code>isdf_metric_omega</code>	0	attenuation width (0: automatic)

columns as N_L nonnegative separable terms; positivity of the expansion preserves the positive semidefiniteness of the selection matrix by the Schur product theorem, and the pivoted factorization retains its unconditional numerical stability. The re-ranking step operates on the Cholesky factor produced by the first pass, so the exactness statement of Eq. (8) holds for any combination of the preceding schemes: the three modifications compose.

For pseudopotentials with augmentation charges (ultrasoft and PAW), the implementation appends atom-centered augmentation channels to the interpolation basis after point selection, ranked by a $\mathbf{q}$ -aware Coulomb norm. A Coulomb-metric point selection thus renders the smooth and augmentation sectors of the auxiliary basis mutually consistent.

III. RESULTS

A. Computational details

We consider bulk silicon in the diamond structure, described by norm-conserving (ONCV), ultrasoft, and PAW pseudopotentials, with Γ -centered $2 \times 2 \times 2$ Monkhorst–Pack sampling for the pseudopotential comparison and symmetry-free $2 \times 2 \times 2$ and $4 \times 4 \times 4$ grids with 60 bands for the production scans [cutoffs and remaining numerical parameters from the run inputs]. Correlation energies are evaluated within the RPA on the imaginary axis using a discrete Lehmann representation at inverse temperature $\beta = 1000 \text{ Ha}^{-1}$. For the norm-conserving calculations, reference integrals are obtained from a Cholesky decomposition of the Coulomb tensor converged to a threshold of 10^{-10} , evaluated through

FIG. 1. [Error in the exchange, Hartree, and RPA correlation energies of Si (norm-conserving pseudopotential) as a function of c_μ , for the conventional selection and for the schemes of Sec. II.]

the identical energy pipeline; the ISDF representation at $c_\mu = 20$ reproduces these reference energies to better than 3×10^{-5} Ha in the exchange and 2×10^{-6} Ha in the RPA correlation energy, validating both the factorization and the error measure. For the augmented pseudopotentials, where the Cholesky route is unavailable, the reference is the ISDF result at $c_\mu = 20$. Errors are reported for the exact-exchange and RPA correlation energies; the Hartree energy, which is available directly from the density at negligible cost and need not be represented by the factorization, is monitored only as a consistency check. As the figure of merit we use c_μ^* , the smallest c_μ at which a given error target is met, obtained by logarithmic interpolation of the error curves.

B. Selection schemes at fixed system size

[Insert final 2x2x2 comparison table/figure: baseline vs the three schemes and compositions; c_μ^* table under both 1 mHa and 0.1 mHa targets; the optimized sigmoid result and its transfer behavior; s -dependence.]

C. Production-scale scans

[4x4x4, 60 bands: error curves for the winning configurations, composition, and recommended defaults; timing of the selection step (IpRerank timer) demonstrating the

$s \times$ overhead claim.]

IV. DISCUSSION AND CONCLUSIONS

Two observations from the numerical results deserve emphasis. First, the appropriate selection metric is observable dependent. The exact Coulomb-metric re-ranking improves the exchange and correlation energies simultaneously, whereas the filtered surrogate improves correlation-type observables at the expense of exchange, whose short-range content the filter deprioritizes. Since the Hartree contribution can be evaluated directly from the density at negligible cost, without recourse to the factorized integrals, the fidelity of the factorization matters precisely for the exchange and dynamically screened contributions; our recommendations are scored accordingly. Second, the compression attainable by improving the selection is bounded by the intrinsic rank of the pair-density manifold: once the selected points resolve that manifold in the Coulomb metric, further gains require compressing the interpolation basis downstream rather than choosing better points. The saturation of the error curves at large c_μ provides a practical indicator of this crossover.

[Finalize after production runs: regimes where the metric matters most (vacuum regions, localized orbitals, metals — the overlap metric is known to overspend in vacuum); recommended defaults (metric choice, s , attenuation); the optimized rank-1 weight; extension of $\bar{v}$ to static screened interactions, where the same re-ranking applies verbatim with $W(\mathbf{G})$ in place of $v(\mathbf{G})$.]

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